

RUSHIK SANJAYBHAI DESAI

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Summary: Software Engineer with 2+ years of software development experience, specializing in **Python (FastAPI, Django, Flask)** and scalable RESTful APIs. Skilled in database optimization (**PostgreSQL, MongoDB, Redis**), cloud technologies (**AWS**), and security best practices (JWT, OAuth). Proven track record of delivering high-performance systems with **microservices, caching, and event-driven architectures**.

Professional Experience

Software Engineer – Protech Consulting (July 2024 – present)

Tech: Python, FastAPI, MongoDB, Redis, Kafka, AWS (EC2, S3, Lambda, CloudFront)

- Increased platform scalability by supporting **5,000+** users within **6 months** through architecting and deploying microservices with FastAPI and PostgreSQL on AWS, resulting in over **98.5%** system uptime and seamless user experience.
- Improved user engagement by **25%** by building real-time notification services with Redis Pub/Sub and WebSockets, resulting in faster interactions and enhanced responsiveness across core application features.
- Enabled consistent **weekly** production releases by taking **end-to-end ownership** of backend infrastructure, containerizing services with Docker, orchestrating workloads with basic Kubernetes, and implementing Jenkins CI/CD pipelines, resulting in faster development cycles and improved deployment reliability.

Software Engineer - Tata Consultancy Services (August 2021 – August 2022)

Tech: React, Python, Django, PostgreSQL, AWS (EC2, S3, RDS, Lambda)

- Streamlined client onboarding and project tracking by building and maintaining internal management portals with Django and PostgreSQL, resulting in improved operational efficiency across **50+** enterprise customers and enhanced user access control for seamless management.
- Reduced manual reporting efforts by **40%** by developing serverless data processing pipelines with AWS Lambda and S3, resulting in automated report generation and faster, real-time insights delivery for improved decision-making.
- Reduced deployment errors by **20%** by implementing CI/CD pipelines with GitHub Actions and AWS CodeDeploy, resulting in faster release cycles, improved testing automation, and more reliable staging deployments.

Software Engineer (Intern) - Isparrow (May 2020 – August 2021)

Tech: Java, Spring Boot, MongoDB, WebSockets

- Handled over **10,000** support tickets monthly by designing and implementing a modular ticketing backend service with Spring Boot and MongoDB, resulting in real-time status updates via WebSocket connections and improved support team efficiency.
 - Reduced average ticket retrieval latency by **35%** and improved API response times by **40%** under load by optimizing database queries and indexing strategies, resulting in faster data access and a more responsive system.
 - Achieved over **95%** unit test coverage by collaborating with frontend and QA teams in Agile sprints to deliver secure RBAC modules, resulting in improved security and reliable feature delivery through comprehensive testing with JUnit and Mockito.
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Technical Skills

- Programming Languages:** Python, JavaScript, Java
 - Frameworks:** Fast API, Django, Node.js, Express.js, React, TensorFlow, Keras
 - Databases:** MySQL, PostgreSQL, MongoDB, Redis
 - Cloud & Platforms:** GitHub, Docker, AWS, Kafka, Jenkins
 - Methodologies:** Agile, Scrum
 - Other:** Data Structures, Algorithms, OOP, Debugging, Problem Solving
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Education

M.S. Computer Science - California State University, San Bernardino (August 2022 – May 2024)

- Relevant Courses:** Software Engineering, Machine Learning, Operating Systems, Artificial Intelligence

B.Tech Computer Science and Engineering - Vellore Institute of Technology (August 2017 – June 2021)

- Relevant Coursework:** Database Management Systems, Network and Communications, Parallel and Distributed Computation, Cyber Security, Natural Language Processing, Operating Systems, Internet and Web Development.
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Projects

Backend MicroServices for MarketPlace Web Application(February 2025 - present)

Tech: NodeJS, ExpressJS, MongoDB, Redis, Kafka

- Developed a microservices-based e-commerce backend, utilizing **MongoDB** and **Kafka** for scalable, fault-tolerant data storage and event-driven communication.
- Implemented real-time inventory management with **Redis**, reducing latency by **40%**, and integrated **Stripe** for payments.

Music Genre Classification with CNN (January 2023 - present)

Tech: Python, TensorFlow/Keras, Librosa, GTZAN, Google Colab

- Processed GTZAN dataset using Librosa to extract Mel-spectrogram features and trained a **CNN** for music genre classification.
- Optimized the model with **Adam** and custom F1-score metrics, deploying the training pipeline on Google Colab with **data augmentation** for improved performance.